

Course Code : 2104CS352**Date** : 07-11-2023**Course Name** : Internet of Things**Duration** : 150 Minutes**Total Marks** : 70**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Recite IoT system architecture levels. **4**
- (B)** Define IoT and its characteristics. **3**

OR

Define Sensors and Actuators with examples.

- (C)** Explain 7-layer IoT architecture with diagram. **7**

OR

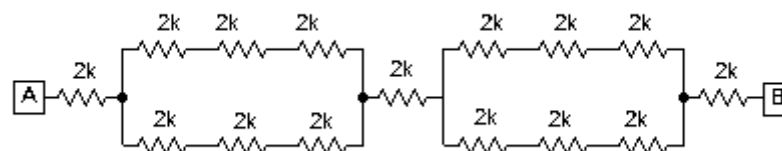
Discuss Enabling technologies and explain Big Data Analytics in detail.

- Q.2 (A)** Define Ohm's law for the Electrical circuit. **4**
- (B)** Write down SI Unit of followings: **3**
- a) Current
 - b) Voltage
 - c) Power
 - d) Resistor
 - e) Capacitor
 - f) Inductor

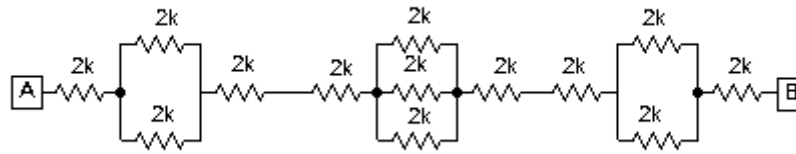
OR

State Kirchhoff's Current Law and Kirchhoff's Voltage Law for electrical circuit.

- (C)** Find the equivalent resistance- R_{AB} between node A and B. **7**

**OR**

Find the equivalent resistance- R_{AB} between node A and B



- Q.3** (A) Define Relay with its diagram and terminals. **4**
 (B) Write an Arduino code to detect object using obstacle sensor. **3**

OR

Write an Arduino code to detect and print room temperature on serial monitor.

- (C) Write an Arduino code to measure distance using Ultrasonic sensor. **7**
 (Speed of Ultrasonic Sound = 0.0345 cm/micro-sec)

OR

Write an Arduino code and draw the circuit to turn on -off Bulb (230 volt).

- Q.4** (A) Define Pulse Width Modulation and explain analog out in Arduino. **4**
 (B) Discuss digital port in Arduino UNO. **3**

OR

Discuss Arduino UNO port for reading and writing analog value.

- (C) Sketch the Arduino UNO board architecture in detail. **7**

OR

Demonstrate all Serial communication function available with Arduino.

- Q.5** (A) Define cloud computing and list it's benefits. **4**
 (B) Discuss a function to read data from Blynk cloud dashboard. **3**

OR

Discuss a function to Update data on the dashboard (Blynk IoT cloud).

- (C) Explain Cloud Computing Services in detail. **7**

OR

Explain Fog computing and compare it with cloud computing.
